Student Assessment Submission and Declaration

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Student declaration I certify that the assignment submission is entirely my own work. I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.			
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ACADEMY MANAGEMENT DATABASE DESIGN

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Conceptual diagram

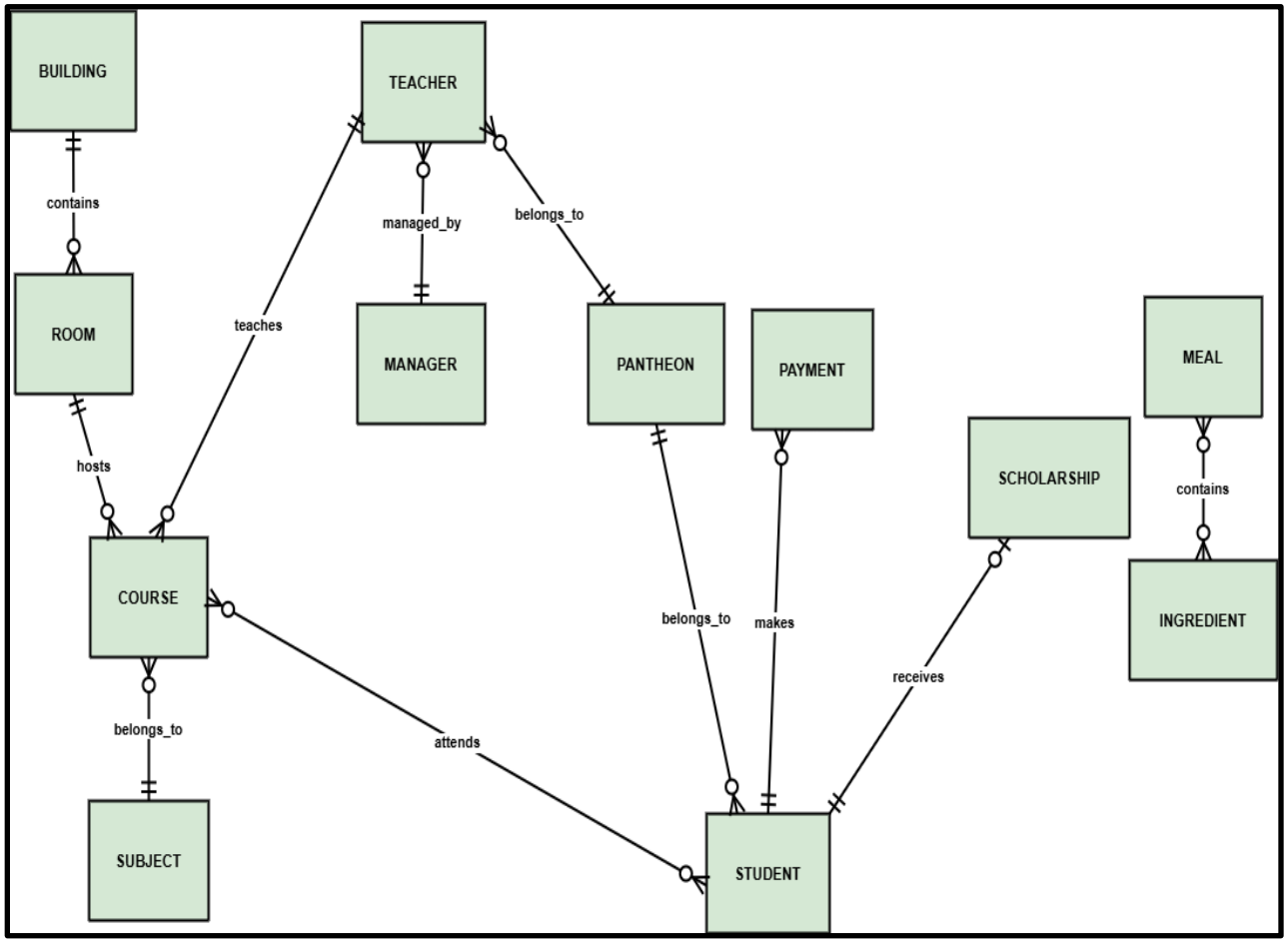

Figure 1: Conceptual ER Diagram

A conceptual ER diagram is a high-level view of the database system that would be built in the academy of villainous opportunity. It determines the key participants of the system, such as Student, Teacher, Manager, Course, Subject, Room, Building, Pantheon, Meal, Ingredient, Payment and Scholarship. The diagram is concentrated on the illustration of the relations between these entities without detailed attributes (Tarigan *et al.* 2024). The courses and students have a many-to-many relationship, whereas the teachers have a one-to-many relationship, suggesting that teachers can teach several courses. Courses are related to a subject, and the courses are in rooms and buildings, respectively. The organizational alignment is that both the students and teachers are attached to a pantheon. Also, payments are made by students, and they can also receive scholarships at will. Ingredients are related to meals in a many-to-many manner. In general, the conceptual diagram offers a simplified illustration of the system requirements derived during interviews, guaranteeing the clarity of relationships and cardinality.

Physical diagram

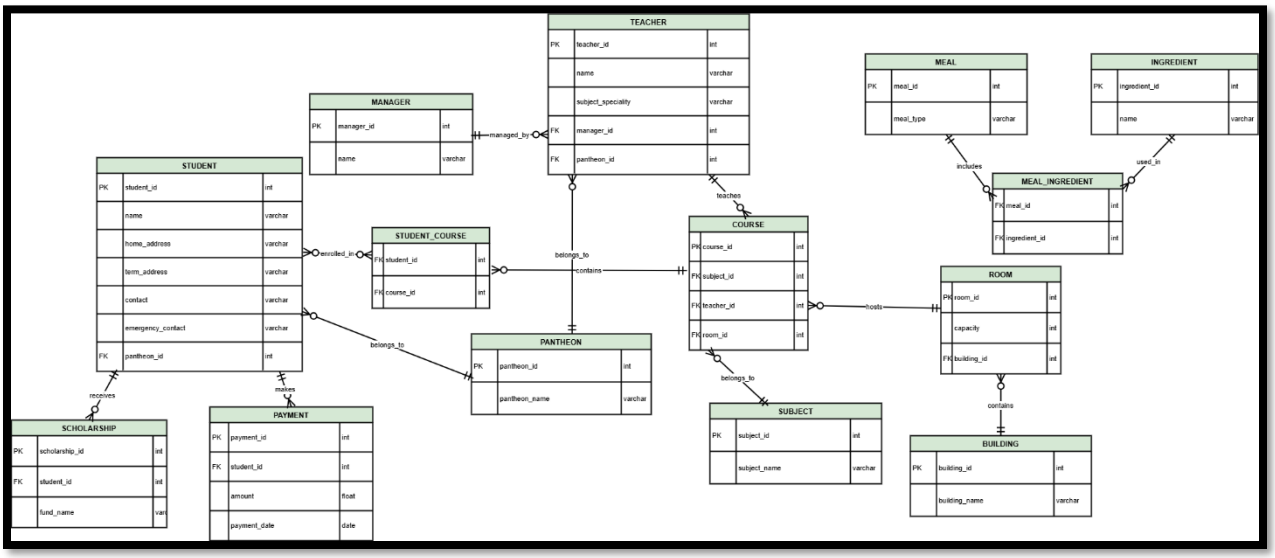

Figure 2: Physical ER Diagram

A physical ER diagram gives a specific representation of the database design through a translation of the conceptual model into the relational tables with attributes, primary keys, and foreign keys. The implementation of each entity is done in the form of a table, e.g. Student, Teacher, Course, Subject, Room and Building with distinct identifiers and data types. Primary keys are used to identify each record uniquely, whereas foreign keys are used to create relationships among tables. Associative tables, like Student Course and Meal Ingredient, are the solution to many-to-many relationships found in the conceptual model. Such tables make sure that they are normalized correctly and that they do not have redundancy. The fact that in the system, tables like Payment and Scholarship are included allows the system to control the financial transactions and sources of funds efficiently. Other constraints that are included in the diagram are the relationships between teachers and managers, as well as room reservations in courses. The design guarantees the integrity of data, scaling and efficient querying of the data; therefore, addressing the operational requirements of the institution.

Short description

The database design is created based on the main requirements of the interview data, which are extracted and converted into a structured model. This is initiated by identifying core entities like students, teachers, courses and facilities, after which the relationships between them are defined. The conceptual diagram is diagrammed in a manner that gives a high-level visualization of the relationships between various components that are involved in the system. This is enhanced by the physical diagram, which adds attributes, a primary key, and a foreign key, which are used to ensure that every table can be identified uniquely and related well (Fatmariani *et al.* 2023). The schema is designed in a Third Normal Form (3NF) because all the non-key attributes are dependent on the primary key only, and all transitive dependencies are eliminated. The details of the managers are included on a different table, instead of being contained in the teacher table, and this minimizes redundancy. Primary keys are selected as unique identifiers, such as student-id and teacher-id, to ensure consistency. The many-to-many relationships, like the one between students and courses, are addressed with the help of junction tables that offer efficient data management.

GitHub link for diagrams:
https://github.com/Junk3r3001/Databases_assessment1/tree/main/Assessment%201%20Database%20model%20with%20annotations

Entity	Primary Key	Key Attributes	Description
Student	student_id	name, home_address, term_address, contact, emergency_contact, pantheon_id	Stores all student personal and academic details
Teacher	teacher_id	name, subject_speciality, manager_id, pantheon_id	Contains teacher information and management hierarchy
Course	course_id	subject_id, teacher_id, room_id	Represents classes assigned to teachers and rooms
Subject	subject_id	subject_name	Defines subjects taught in the academy
Room	room_id	capacity, building_id	Stores room details and capacity constraints
Building	building_id	building_name	Represents physical infrastructure
Pantheon	pantheon_id	pantheon_name	Groups students and teachers by alignment

Table 1: Core Entities and Key Attributes

Relationship / Table	Type	Connected Entities	Description
Student_Course	Many-to-Many	Student ↔ Course	Tracks student enrolment in multiple courses
Meal_Ingredient	Many-to-Many	Meal ↔ Ingredient	Stores ingredients used in each meal
Payment	One-to-Many	Student → Payment	Records all student fee transactions
Scholarship	Optional (1:1)	Student → Scholarship	Stores scholarship or bursary information
Teacher_Manager	Many-to-One	Teacher → Manager	Defines reporting hierarchy
Room_Course	One-to-Many	Room → Course	Assigns rooms to courses

Table 2: Relationships and Associative Entities

References

Fatmariani, F., Putri, M.P. and Apriliani, M., 2023. Implementation of CV Metha Developing Palembang Goods Ordering System. Sinkron: jurnal dan penelitian teknik informatika, 7(1), pp.442-450.

Tarigan, F.J., Br Bukit, R. and Erwin, K., 2024. Analysis of Factors Affecting the Successful Implementation of Accounting Information Systems with Availability as a Moderating Variable in Online Queues with Entity Relationship Diagram Concepts in North Sumatra Provincial Type A Hospitals. International Journal of Research and Review, 11(6), pp.884-897.